Mastering Generative AI & LLM Engineering
From Foundations to Applications
Edited by Dr. Divyas Shukla
in association with
iHUB DivyaSampark, IIT Roorkee & RBPL

LLM Engineering

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From Foundations to Applications

Agenda :

- prompt engineering
- memory handling
- cost optimization
- security and data safety
- scalability and performance
- and dealing with real-world issues like hallucinations and prompt injections

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Stages of Development :

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Stages of Development :

1. Planning Stage

- Decide: self-host LM or API?
- Choose the right models
- Understand tokens & pricing
- Map flows with activity diagrams

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Stages of Development :

2. Prototype (Hands-On)

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- Configure GPT settings
- Learn Prompt Engineering
- Build app with Streamlit
- Create & deploy working prototype
- Push to GitHub + Streamlit Cloud

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Stages of Development :

3. Advanced Development

- System architecture & LM components
- Handle inconsistencies & hallucinations
- Prevent prompt injections
- Optimize cost & scalability

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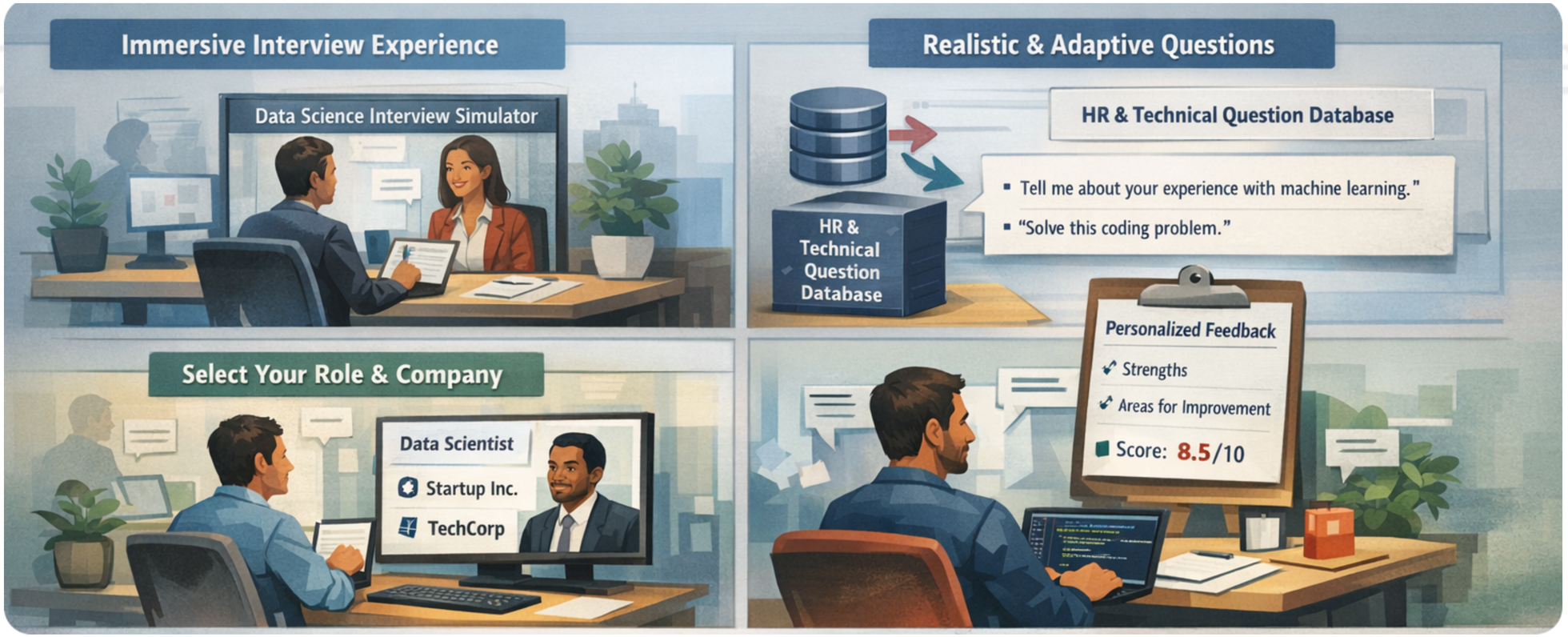

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Immersive Interview Experience
Data Science Interview Simulator
Realistic & Adaptive Questions
HR & Technical Question Database
HR & Technical Question Database
Tell me about your experience with machine learning."
"Solve this coding problem."
Personalized Feedback
Strengths
Areas for Improvement
Score: 8.5/10
Select Your Role & Company
Data Scientist
Startup Inc.
TechCorp
HR Interview Simulation
"How do you handle workplace conflict?"
Behavioral
Situational
Technical Interview Simulation
Write a SQL query to analyze this dataset."
Coding
Algorithms
Key Requirements Recap:
Realistic Interview Simulations
Choose Position & Company
Provide Experience & Skills
Detailed Feedback & Scoring

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Planning Stage

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Things to Consider

- LLMs : self-host or use API ?
- Tokens : Usage, Pricing etc.
- Cost Optimization
- Models to consider (OpenAI’s GPT, BERT , Gemini etc.)

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Model Integration

1. Self-Hosting

- Running model on your own servers or cloud
- Full control and customization, better privacy
- GPUs needed and so is storage (a lot of it)

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Model Integration

2. Use API

- Source an external model
- No complex hardware and infrastructure
- Easy and quick to integrate and scale
- Less downtime, security risks or policy changes

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Choosing the right LLM

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Choosing the right LLMs

1. Open - source

- Publicly available to fine-tune, tweak, fork or plug them into your system. (Eg : BERT, GPT-2, Mistral, LLaMA etc.)
- More parameters usually mean the model can learn more complex size
- Bigger models means more compute and memory requirements
- Data is always the King.
- Check Hugging Face Open LLM Leaderboard

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Choosing the right LLMs

2. Closed - source

- Not publicly available
- Need to be accessed through APIs
- Scalable, Easy to Use and Deep Learning expertise not needed
- Less controllable, can’t modify or retrain it

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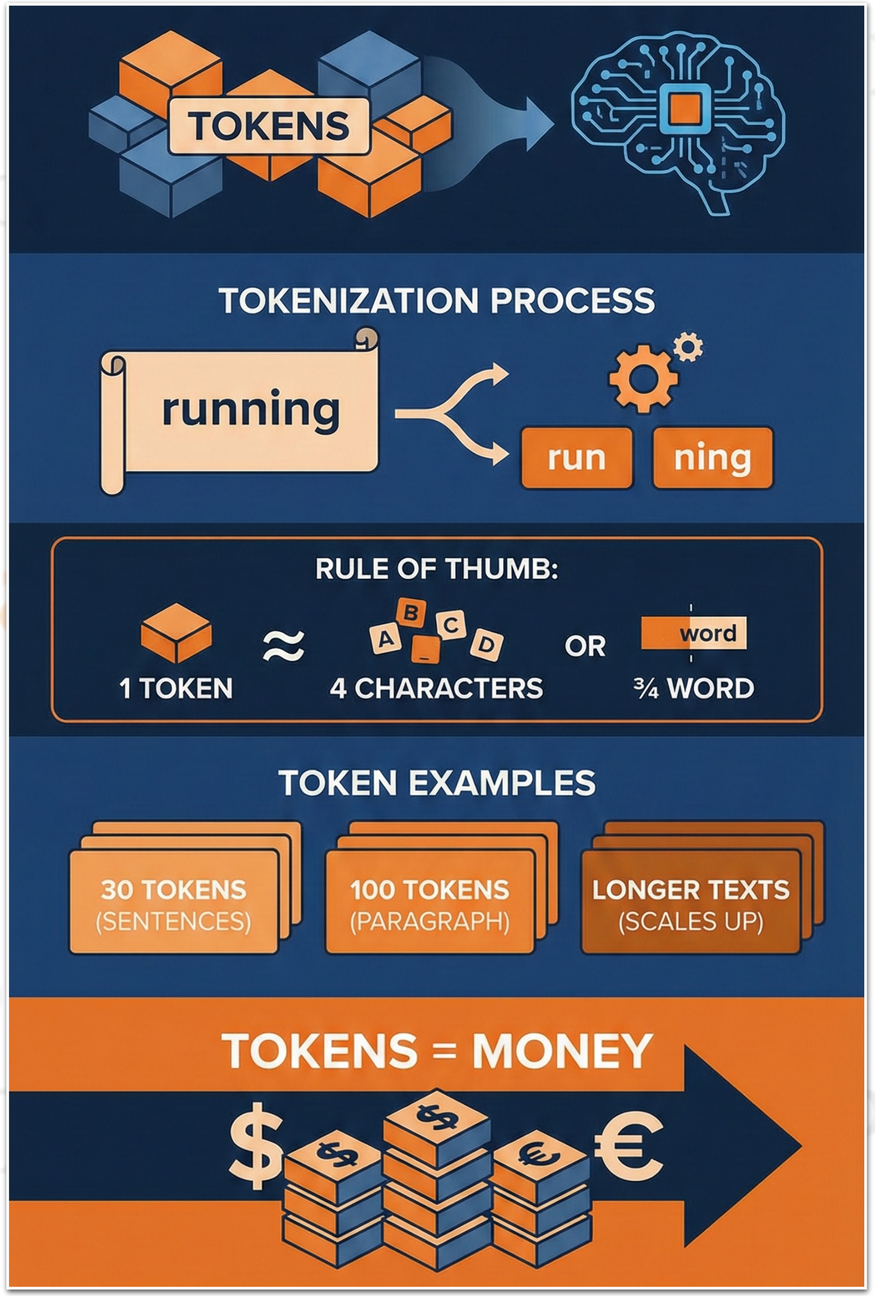
Tokens

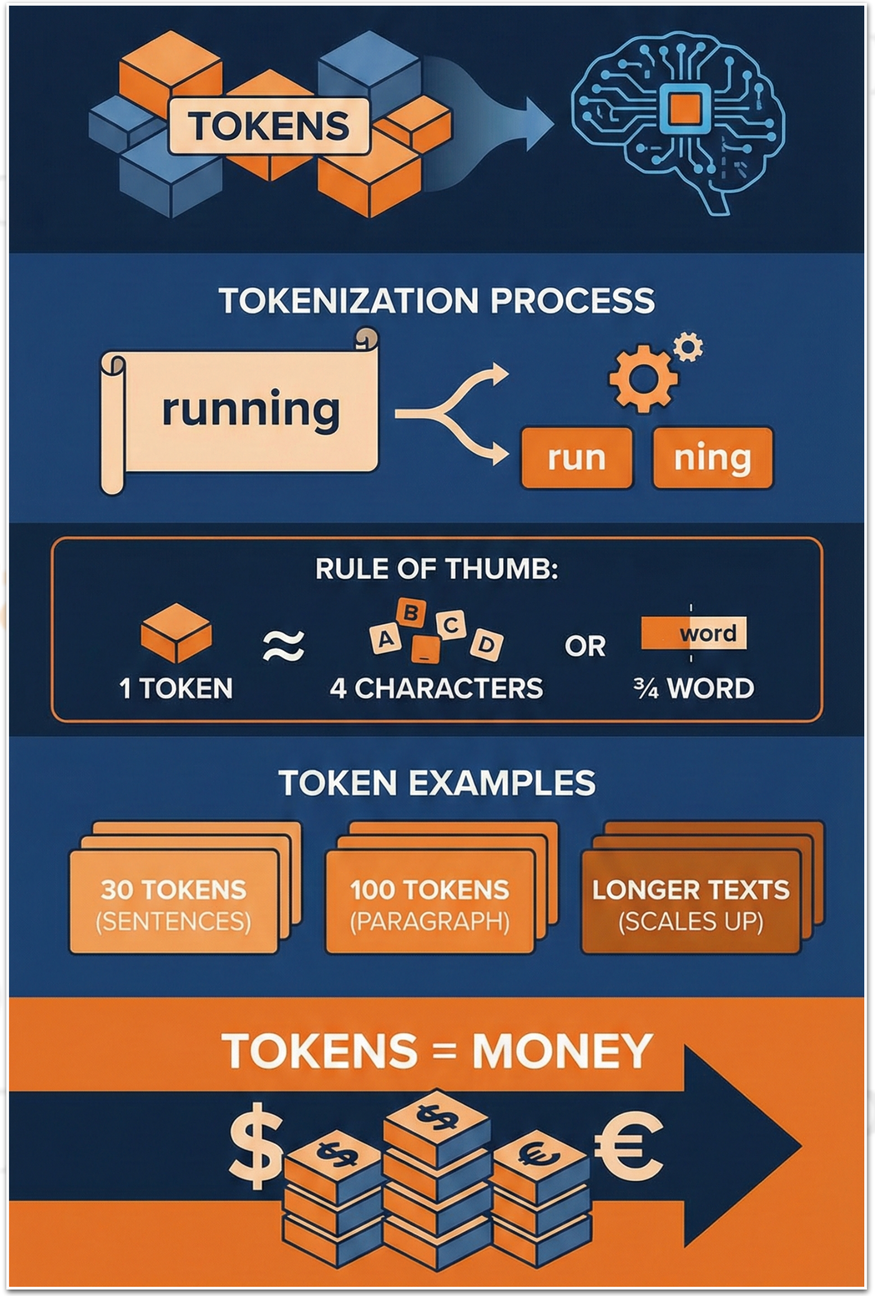
Tokens

- Tiny building blocks for LLMs
- Could be full words, parts of words or even single characters
- one token ≈ 4 characters, or about ¾ of a word.



Tokens

- Context Window : How much text the model can ‘see’ at once.
- Token Limit : Total number of tokens allowed in a single request.



TOKEN LIMIT SPLIT: 4000 TOTAL

How Prompt vs. Response Allocation Impacts Output

BALANCED SPLIT: BEST CASE

Plenty of room for both clear instructions and a solid answer

SHORT PROMPT: LONG OUTPUT

Great space for detailed generation. Perfect for long outputs.

HUGE PROMPT: LIMITED OUTPUT

Limited response space. Leads to shorter replies or summaries.

WORST CASE: BARELY SPACE

Barely space to respond. Results in cut-off, rushed, or half-finished outputs.

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Token Pricing

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Token Pricing

- Price vs. Performance: Switching to smaller models (e.g., GPT-3.5 Turbo) can be 10x cheaper but sacrifices some output quality.
- Why APIs First: APIs remove the need for expensive hardware or complex ML Ops, allowing you to "plug and build."
- Closed Source Benefits: Options like OpenAI offer reliability, support, and automatic updates out of the box.
- Lean Strategy: Start with APIs to validate your idea cheaply; switch to self-hosted open-source models later to optimize costs as you scale.

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Designing the System Prompt

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Designing the System Prompt

- System Messages : Developers instructions to the model to tell it how to behave
- User Messages : Inputs from End users
- AI Messages : Model’s response

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Designing the System Prompt

For our interview simulator :

- System Message : to define how the interviewer behaves
- User Messages : for the candidate’s answers
- AI Messages : for the interviewer’s questions and guidance.

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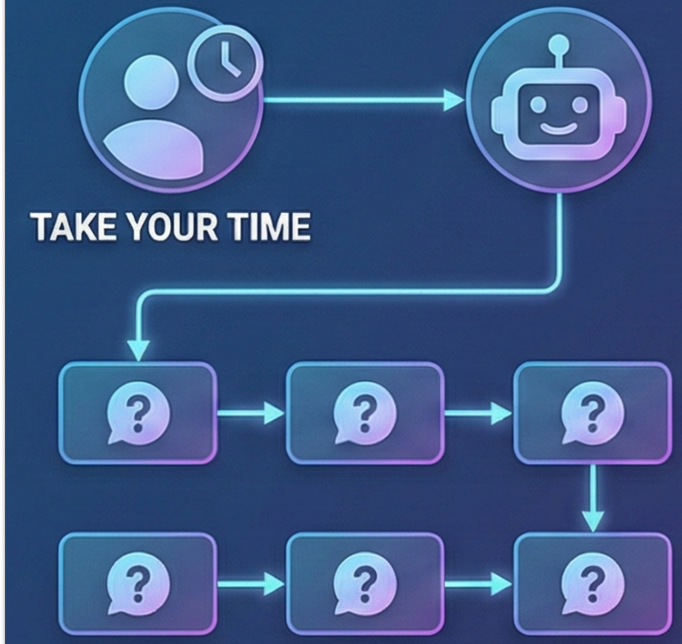
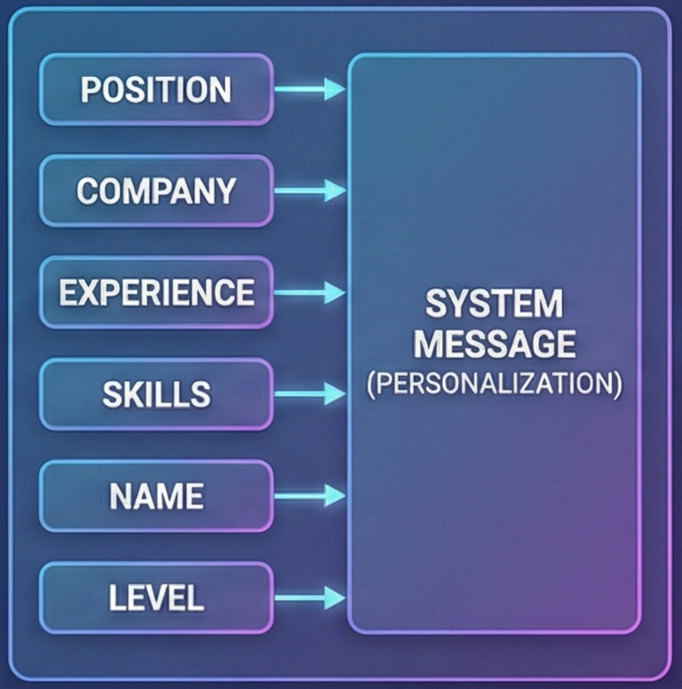
INTERVIEW STRUCTURE (SELF-PACED)



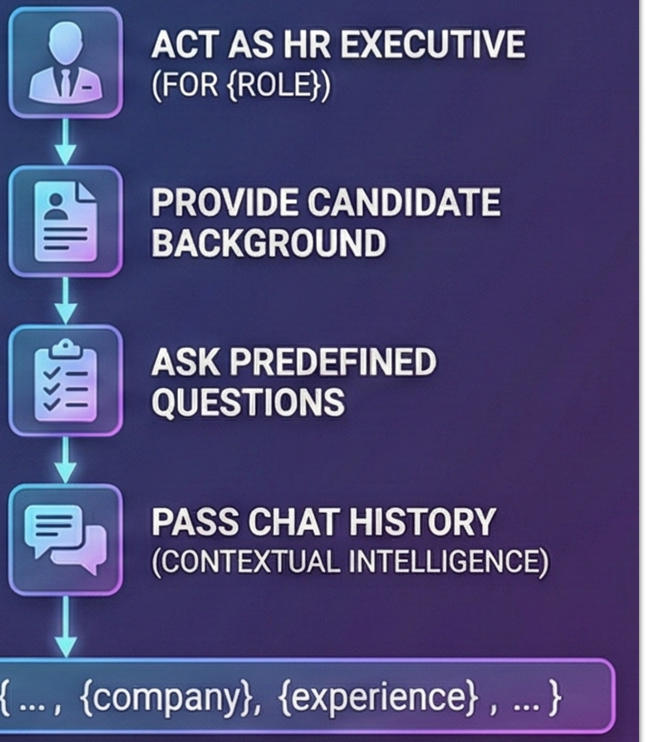
Simulates 30-40 min, no pressure.

PROMPT INPUTS & SYSTEM MESSAGE



User-specific data shapes the persona.

PROMPT TEMPLATE FLOW (PLACEHOLDERS)



PLACEHOLDERS FOR DYNAMIC FILLING
(UPCOMING LESSONS)

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"You are an HR executive at **{Company}**, interviewing a candidate named **{Name}** for the position of **{Position}** at the **{Level}** level.

The candidate has the following experience: **{Experience}**

The candidate has the following skills: **{Skills}**

Use the following questions to create at least **two** of your interview questions: **{Questions}**

Chat history for context: **{Chat_history}**

Conduct the interview by asking **six questions total**, one at a time, and wait for the candidate's response after each question.

After all six questions and answers are completed, evaluate the candidate's performance and provide:

- An overall score from **1 to 10**
- A clear hiring recommendation based on:
  - **1–4**: Unlikely to be considered
  - **5–6**: Slim chance of being hired
  - **7–8**: Promising candidate
  - **9–10**: Top contender
- Detailed feedback
- A section outlining **areas for improvement"**

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Reduced Complexity but..

- Limited Flexibility
- Difficulty handling complex or diverse tasks
- Quickly hitting token limits
- Producing off-topic or incorrect outputs

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The Alternative: Two Model Approach

Interview Model

Conducts Interview

- Asks Questions
- Manages Flow

Chat History

Evaluation Model

Analyzes & Scores

- Focuses on Feedback
- Provides Scores

Better Feedback | **More Stable Process** | **Lower Cost Evaluation Model**

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The Alternative: Two Model Approach

Interview Model

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Better Feedback | **More Stable Process** | **Lower Cost Evaluation Model**

The Challenges

Increased Complexity

Higher Resource Usage

More Coordination Needed

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Primary Key

Unique ID (No Duplicates, No Nulls)

ID	Name	Email
101	Alice	alice@example.com
102	Bob	bob@example.com
103	Charlie	charlie@example.com

Foreign Key

Reference to Another Table

Orders

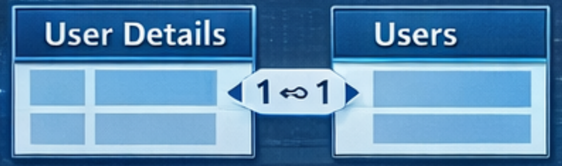
Order ID	User ID
201	101
202	103

Users

ID	Name
101	

Types of Relationships

One-to-One

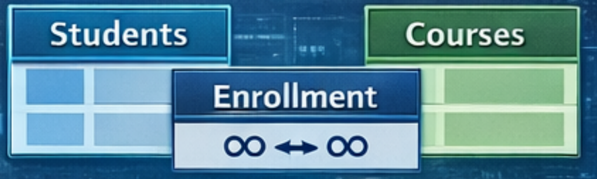


1 Row ↔ 1 Row

One-to-Many

1 Row → Many Rows

Many-to-Many



Many Rows ↔ Many Rows
(Join Table)

iNeuron SQL

Positions Table

Position ID	Position Name
1	Instructor
2	Mentor
3	Admin

Position ID *(Primary Key)*

Stores course roles.

Industries Table

Industry ID	Industry Name
1	Data Science
2	AI & Machine Learning
3	Analytics

Industry ID *(Primary Key)*

Lists data science industries.

Companies Table

Company ID	Company Name	Industry ID
101	DataCorp	1
102	Tech Innovations	2
103	InsightWorks	1

Industry ID *(Foreign Key)*

Links companies to industries.

Questions Table

Question ID	Question Text	Position ID	Company ID
201	"SQL Basics?"	1	101
202	"Machine Learning Tips?"	2	102

Position ID & Company ID *(Foreign Keys)*

Holds *job-related questions.*

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The Architecture

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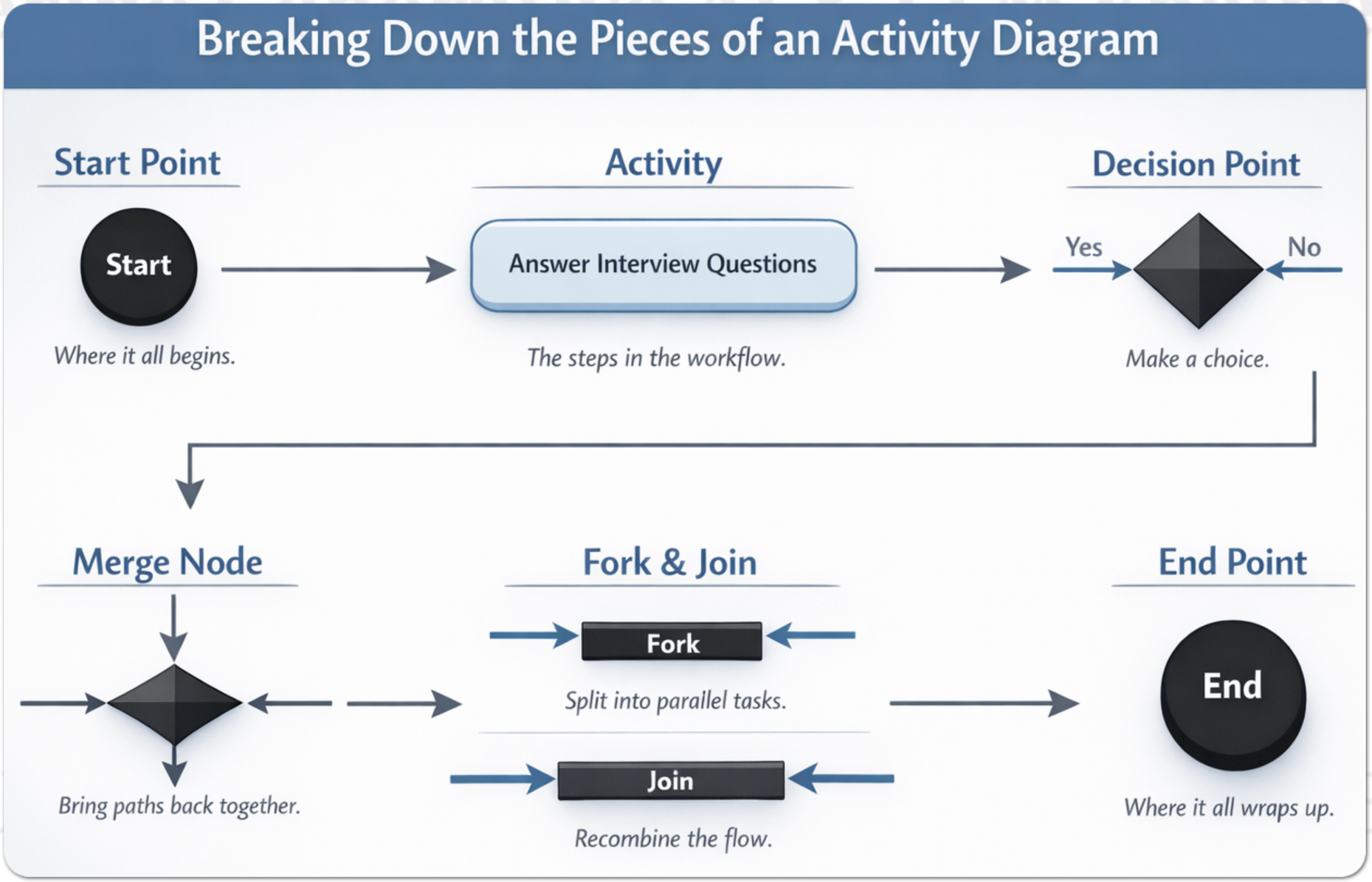
Breaking Down the Pieces of an Activity Diagram



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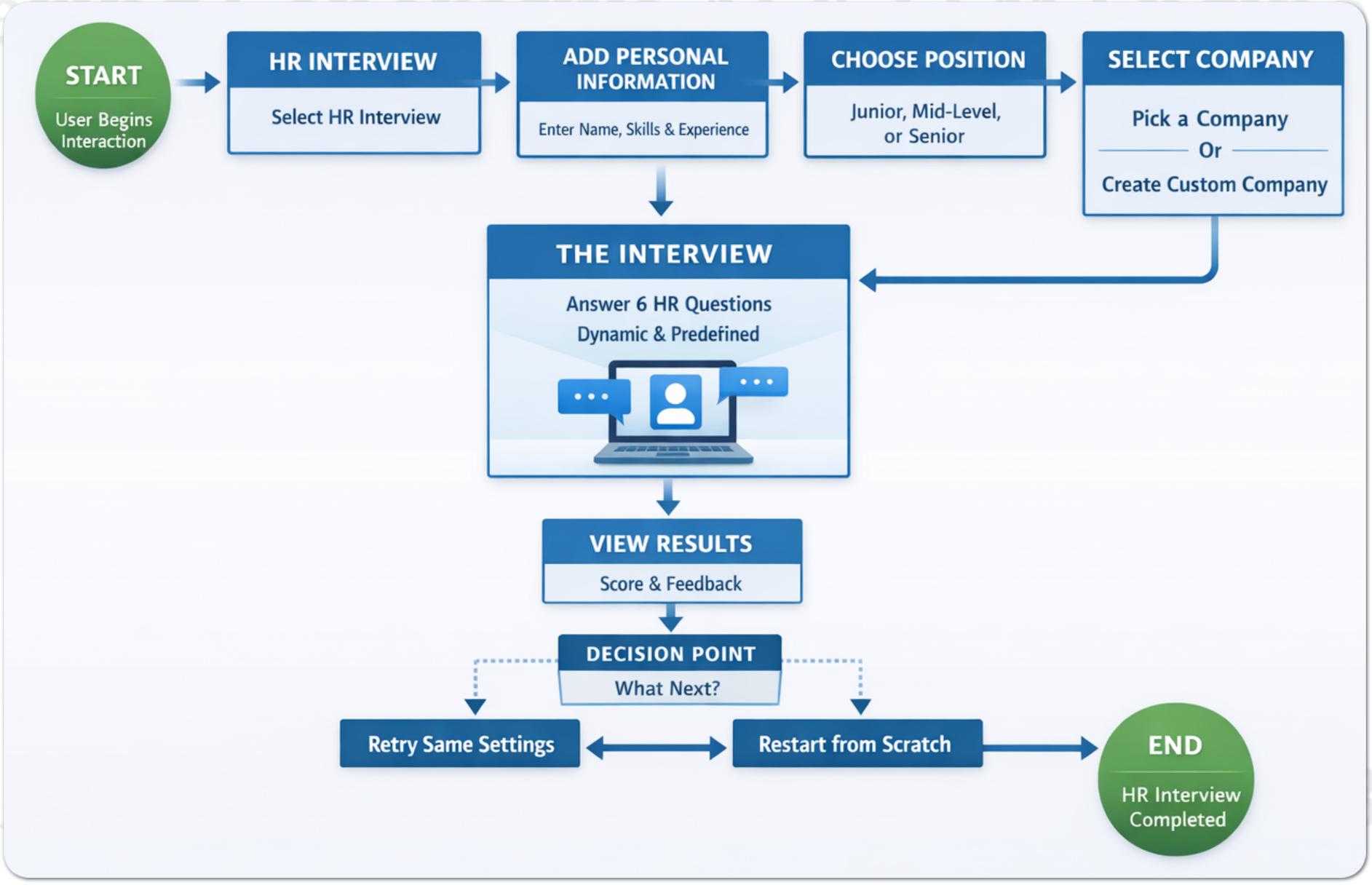

iHUB PL

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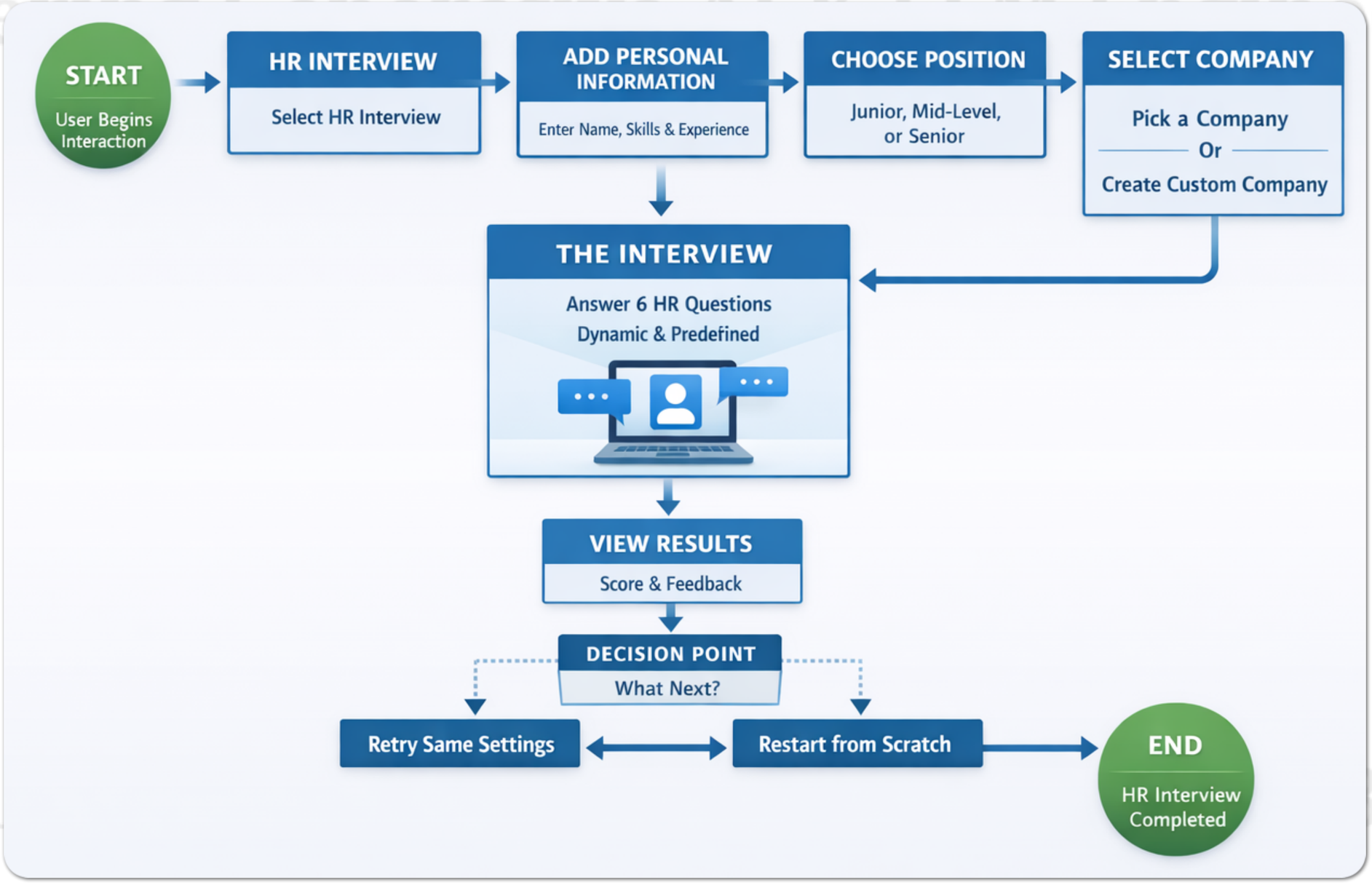
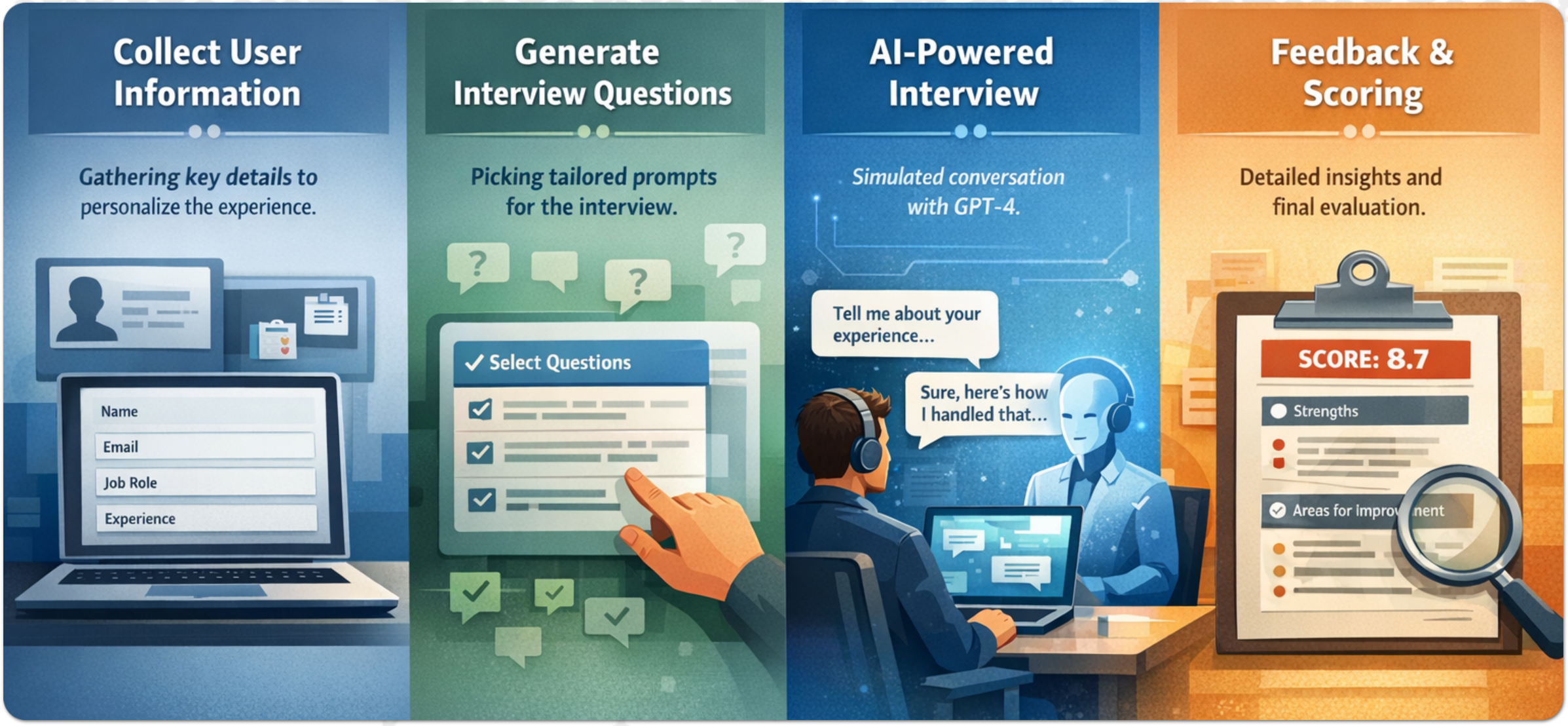


iHUB PL


Collect User Information
Gathering key details to personalize the experience.
Name
Email
Job Role
Experience
Generate Interview Questions
Picking tailored prompts for the interview.
Select Questions
AI-Powered Interview
Simulated conversation with GPT-4.
Tell me about your experience...
Sure, here's how I handled that...
Feedback & Scoring
Detailed insights and final evaluation.
SCORE: 8.7
Strengths
Areas for improvement

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The Prototype

Exploring GPT Models
The Playground
Enter Your Prompt...
Temperature
Max Tokens
Building with Streamlit
Easy Python Apps
Connecting It All
OpenAI + Streamlit
Interview Tool Prototype
Question:
Answer:
OpenAI
Deploying Your App
GitHub & Deployment
Upload to GitHub
— Now Live! —
APP DEPLOYED!

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https://platform.openai.com/playground

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Optimizing Models

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Temperature and Top P for Different Use Cases

USE CASE	TEMPERATURE	TOP-P	DESCRIPTION
Creative tasks	1.1	1	Encourages experimental and creative outputs with less constraint.
Writing tasks	0.3	0.5	Balances precision and creativity for effective text enhancement.
Coding tasks	0.3	0.3	Ensures reliable, syntactically correct, and focused code generation.
Data Analysis & Scripting	0.2	0.1	Produces highly accurate, deterministic, and efficient data analysis scripts.
Transformation tasks	0.3	0.5	Balances accuracy and flexibility for adaptable, consistent outputs.
Chatbot Responses	1	1	Generates varied, engaging, and human-like interactions.

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Prompt Engineering

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“It’s the art of designing your prompts to get desired output”

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Best Practices

1. Be Clear and Concise

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Best Practices

1. Be Clear and Concise
2. Ask for Structured responses

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Best Practices

1. Be Clear and Concise
2. Ask for Structured responses
3. Few-shot prompting

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Best Practices

1. Be Clear and Concise
2. Ask for Structured responses
3. Few-shot prompting
4. Breaking tasks into steps

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Best Practices

1. Be Clear and Concise
2. Ask for Structured responses
3. Few-shot prompting
4. Breaking tasks into steps
5. Set strict rules

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Best Practices

1. Be Clear and Concise
2. Ask for Structured responses
3. Few-shot prompting
4. Breaking tasks into steps
5. Set strict rules
6. Don’t put everything in one prompt. Split it up.

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Best Practices

1. Be Clear and Concise
2. Ask for Structured responses
3. Few-shot prompting
4. Breaking tasks into steps
5. Set strict rules
6. Don’t put everything in one prompt. Split it up.
7. Balance control and creativity

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